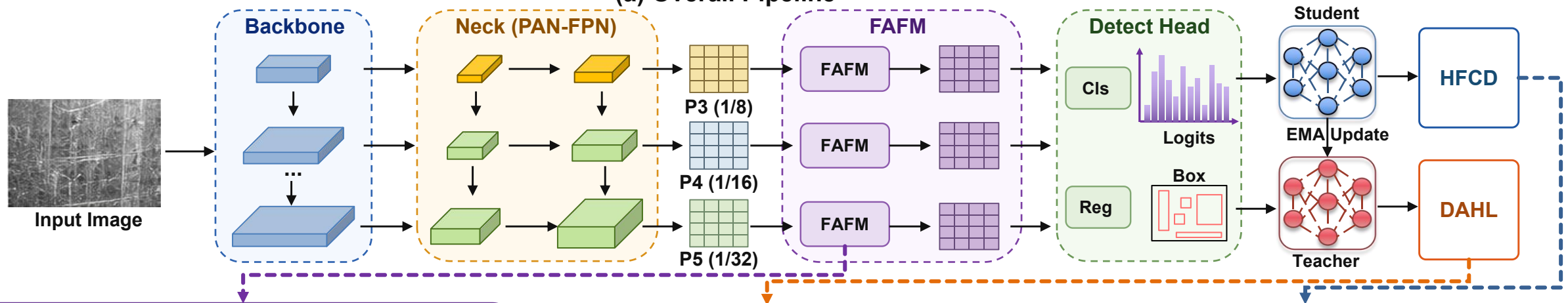
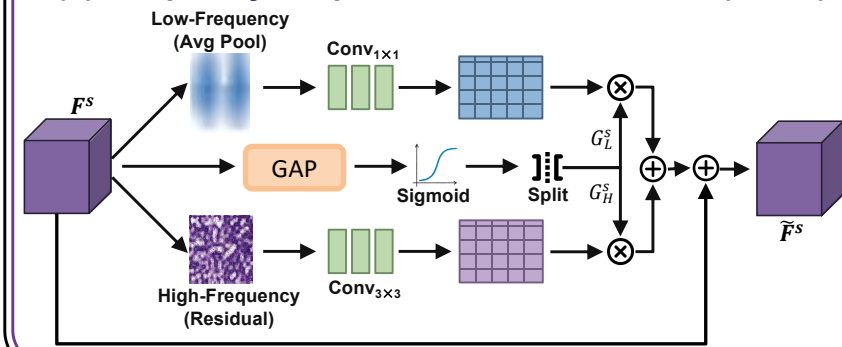


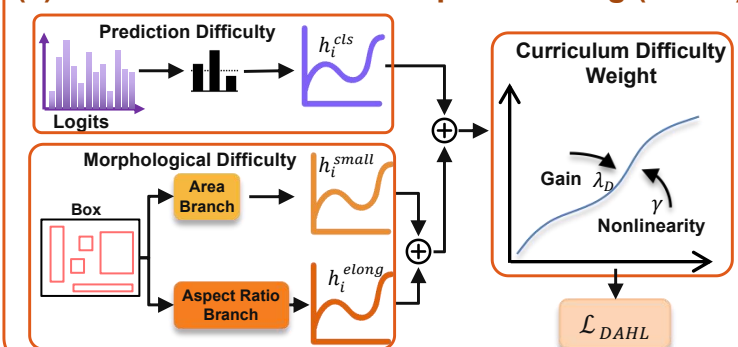
(a) Overall Pipeline



(b) Frequency-adaptive Feature Modulation (FAFM)



(c) Defect-aware Hard-example Learning (DAHL)



(d) Hardness-guided Frequency Consistency Distillation (HFCD)

